



MTH204 : Symmetry (2024-25)

Tutorial 06 (March 22, 2025)

1. Let G be a group. Consider the function $G \times G \rightarrow G$ defined by $(g, x) \mapsto gx$.
 - (a) Show that this function an action of G on itself.
 - (b) Count the number of orbits for this action.
 - (c) Verify Burnside's formula for this action.
2. Let $[1..n] := \{1, 2, \dots, n\}$ and $r \leq n$. Let $a_1, a_2, \dots, a_r$ be distinct elements $[1..n]$, and $\tau = (a_1 \ a_2 \ \dots \ a_r) \in S_n$. Show that $\sigma\tau\sigma^{-1} = (\sigma(a_1) \ \sigma(a_2) \ \dots \ \sigma(a_r))$ for every $\sigma \in S_n$.
3. For a permutation

$$\tau = (a_{11} \ a_{12} \ \dots \ a_{1r_1})(a_{21} \ \dots \ a_{2r_2})(a_{31} \ \dots \ a_{3r_3}) \dots (a_{s1} \ \dots \ a_{sr_s}) \in S_n$$

where $\{a_{11}, \dots, a_{1r_1}, a_{21}, \dots, a_{2r_2}, \dots, a_{s1}, \dots, a_{sr_s}\} = [1..n]$; $r_1 \geq r_2 \geq \dots \geq r_s$; and $r_1 + r_2 + \dots + r_s = n$ is called the *partition type* of τ . Show that τ and τ' are in the same orbit with respect to conjugation action of S_n on itself, if and only if they have the same partition type.

4. Consider $G = \text{sym}(P_4)$, the group of symmetries of a square. For the conjugation action of G on itself, count the number of orbits. Verify Burnside's formula for this action.
5. How many coloured tetradral toys can be made using blue and red colors? Colouring is to be done on faces of tetrahedrons.
6. How many coloured cubic toys can be made using blue and red colors? Colouring is to be done on faces of cube. What if the colouring is to be done on the edges of the cube?